POLLINATION AND FERTILIZATION

Lets go.

📝 Part 1: In-Chapter Point-Wise Notes

🐝 5.1 What is Pollination?

- **Pollination:** The process of transfer of pollen grains from the anther to the stigma.
- The anthers (male organs) produce a powdery material called pollen, which consists of particles called pollen grains.
- Each pollen grain contains **two sperm nuclei** that participate in reproduction.
- Pollination must strictly occur between plants of the **same species**. Only the pollen of the same species will survive and grow further; the rest perish.
- **Three Principal Ways of Pollination:**
    1. **Autogamy (Self-pollination):** Pollen of the same flower falls on its own stigma (*auto*: self, *gamy*: marriage).
    2. **Geitonogamy (Self-pollination):** Pollen of another flower of the same plant falls on the stigma (*geitono*: neighbouring).
    3. **Allogamy (Cross-pollination):** Pollen of a flower of another plant of the same species falls on the stigma (*allo*: other).

🔁 5.1.1 & 5.1.2 Self-Pollination vs. Cross-Pollination

- **Cleistogamous flowers:** Minute bisexual flowers that do not open at all, keeping reproductive structures hidden to facilitate self-pollination ➡️ *(Example: Pea, pansy).*
- **Chasmogamous flowers:** Large flowers with open petals and exposed reproductive structures that mainly undergo cross-pollination ➡️ *(Example: Oxalis, Hibiscus).*

- **Conditions for Self-Pollination:** Requires **Bisexuality** (both male and female organs present) and **Homogamy** (anther and stigma mature at the same time).

Parameter 📏	**Self-Pollination (Autogamy / Geitonogamy)** 🌸	**Cross-Pollination (Allogamy)** 🐝
Pollen Transfer	From anther to stigma of the same flower or another flower of the same plant.	From anther to stigma of another flower of a different plant of the same species.
External Agent	No external agency or agent is required.	An external agent (wind, water, insect) is always required.
Maturation	Male and female parts mature at the same time.	Anthers and stigma mature at different times.
Flower State	Can occur even when the flower is closed.	Can occur only when the flower is open.
Genetic Outcome	Preserves parental characters (no new varieties).	Does not preserve parental characters; new varieties can be produced.
Offspring Vigor	Seeds give rise to less vigorous offspring; weakens the variety over generations.	Offsprings are healthier and better adapted to environmental changes.
Economy	Economical: Less pollen wasted, no need for large	Uneconomical: High pollen wastage; plant must invest in

Parameter 📏	Self-Pollination (Autogamy / Geitonogamy) 🌸	Cross-Pollination (Allogamy) 🐝
	showy petals, scent, or nectar.	large flowers, scent, and nectar.

🌿 Nature's Devices to Favour Cross-Pollination

- **1. Unisexuality:** Flowers are either male or female.
    - ➡️ *(Example: Palms and papaya have separate male and female trees).*
    - ➡️ *(Example: Cucumber and gourd bear both male and female flowers on the same plant).*
- **2. Dichogamy:** Different timings of maturation of androecium and gynoecium.
    - **Protandry:** Anthers mature earlier than the stigma ➡️ *(Example: Lady finger, sweet pea, Salvia, sunflower).*
    - **Protogyny:** Stigma matures earlier than the anthers ➡️ *(Example: Custard apple, peepal).*
- **3. Self-sterility:** Pollen from a flower fails to undergo further growth on the stigma of the same flower ➡️ *(Example: Ray florets of sunflower, orchids).*
- **4. Herkogamy:** Mechanical or structural barriers, such as a hood covering the stigma ➡️ *(Example: Pansy, Iris).*
- **5. Heterostyly:** Stigma and anthers grow at different heights ➡️ *(Example: Primrose, Oxalis).*

🌬️🦋 5.1.3 Agents of Cross-Pollination

- **Insect-pollinated (Entomophilous):** (*entomon*: insect, *phile*: affinity).
    - Flowers are large, brightly coloured, emit scent, and produce nectar.

    - Pollen grains are sticky or spiny.
    - Stigma is sticky and generally does not hang out.
    - Small individual flowers group into clusters to become conspicuous ➡️ *(Example: Dahlia).*
- **Wind-pollinated (Anemophilous):** (*anemo*: wind, measured by an *anemometer*).
    - Flowers are small, dull green (not brightly coloured), and produce no scent or nectar.
    - Stamens are long and hang out ➡️ *(Example: Maize).*
    - Anthers are large and loosely attached (**versatile anthers**).
    - Pollen is produced in huge quantities; grains are light, dry, and smooth.
    - Stigmas are feathery and hang out to trap pollen.
- **Water-pollinated (Hydrophilous):** (*hydro*: water).
    - Found only in aquatic plants.
    - Pollen grains have a specific gravity almost equal to water so they float ➡️ *(Example: Hydrilla).*
    - Male flowers may float to the surface to meet female flowers ➡️ *(Example: Vallisneria).*
- **Bird-pollinated (Ornithophily):** (*ornitho*: bird) ➡️ *(Example: Bignonia, Canna).*
- **Bat-pollinated (Chiropterophily):** Pollination carried out specifically by bats.

📝 **Part 1: In-Chapter Point-Wise Notes**

🆚 **Table 5.2: Wind-Pollinated vs. Insect-Pollinated Flowers**

Parameter 📏	Wind-Pollinated Flowers 🌾	Insect-Pollinated Flowers 🐝
Flower Appearance	Flowers are unisexual, dull coloured, without fragrance and nectar.	Flowers are brightly coloured, fragrant and secrete nectar.
Pollen Quantity	Pollen grains are produced in large numbers, some go waste.	Less pollen grains are produced because they are transported mechanically by insects.
Pollen Structure	Pollen grains are small, light and smooth.	Pollen grains are larger, sticky and spiny.
Stigma	Stigma is long and hangs out of the petals, feathery or sticky.	Stigma often deep in corolla and small.
Stamens	Stamens long and protrude above petals.	Stamens may be within corolla tube.
Inline Examples	➡️ *(Example: Maize, grass, rice, wheat).*	➡️ *(Example: China rose, Salvia, pea, sunflower).*

🐘 Special Pollination Cases & Techniques

- **Elephophily:** Pollination facilitated by elephants.
    - ➡️ *(Example: Found in Rafflesia, whose very large flowers are at ground level).*
    - **Mechanism:** Pollen grains get stuck to the feet of elephants and are carried to another flower when trampled.
- **Artificial Pollination (Artificial Crossing):** When man himself transfers pollen to the stigma to evolve new varieties.

- **Emasculation:** Removing the anthers in young flowers.
- **Bagging:** Covering these flowers with plastic bags before pollinating with the desired variety.

🌸 5.2 Specific Examples of Pollination

- **Insect-Pollinated Sweet Pea:** The bee thrusts its long tongue in search of nectar.
  - The 'wing petals' along with the 'keel petal' are forced down.
  - Stamens and stigma touch the underside of the insect's body, transferring pollen.
- **Wind-Pollinated Maize:** Pollen from outwardly hanging anthers in the tassel (male inflorescence) is blown away easily.
  - Falls on the feathery stigmas of the female flowers (silk/cob), which provide a large surface.
  - **Protandry:** The male flowers mature earlier than the female ones, reducing chances for self-pollination.

🚀 5.3 Fertilization

- **Fertilisation** is the union/fusion of the nuclei of male and female gametes.
- The pollen grain produces **two male gametes**, and the ovule inside the ovary produces **the female gamete**.

1. **The Pollen Grain (Male)**
   - A mature pollen grain has a double wall: the outer **exine** and the inner **intine**.
   - Its nucleus divides into a small **tube nucleus** and a **generative (male) nucleus** before pollination.
2. **The Ovule (Female)**
   - The inner part of the ovary destined to become the seed (while the ovary becomes the fruit).
   - **Integuments:** One or two protective coverings surrounding the ovule.

- **Micropyle:** A small opening left by the integuments at one end for the entry of the pollen tube.
- **Nucellus:** A mass of food-laden cells enclosed by the integuments.
- **Embryo Sac:** Located further inside the nucellus. It contains **7 cells and 8 nuclei** (3+3+1 layout):
    - **Micropylar end:** 3 cells (one **egg cell** and two **synergids**).
    - **Opposite end:** 3 cells called **antipodal cells**.
    - **Center:** 1 large **central cell** containing two nuclei (called **polar nuclei**) fused together.

🌱 Germination of Pollen Grain & Double Fertilization

- **Germination Trigger:** Stimulated by the secretion of sugars by the stigma.
- A pollen tube grows out through a point in the exine, carrying the generative nucleus and tube nucleus at its tip.
- The generative nucleus divides into **two sperm nuclei (two male gametes)**.
- The tube nucleus directs growth down to the ovary and then disintegrates.
- **The Double Fertilization Process:**
    1. **Syngamy:** One sperm nucleus fuses with the egg cell nucleus to form the **zygote**.
    2. **Triple Fusion:** The second sperm nucleus fuses with the two polar nuclei in the central cell (3 nuclei fuse together) to produce the **endosperm nucleus**.
- Because two fertilizations occur simultaneously, the entire process is termed **double fertilisation**.

🍎 Fate of Floral Parts After Fertilization

- After fertilization, the petals, stamens, style, and stigma wither and generally fall off.

- **Calyx (Sepals):** May fall off, or remain intact in a dried and shrivelled form ➡️ *(Example: Apple and guava show dried sepals)*. In some cases, the green calyx persists ➡️ *(Example: Brinjal)*.

Pre-Fertilization Part 📌	**What it Becomes (Post-Fertilization)** 🍏
Ovary	Fruit
Ovary wall	Pericarp (Fruit wall - can be dry/hard or fleshy)
Ovule	Seed
Placenta	Stalk of the seed
Outer integument	Testa (Outer seed coat)
Inner integument	Tegmen (Inner seed coat)
Secondary (Polar) nucleus	Endosperm
Egg cell (via Syngamy)	Embryo
Synergids & Antipodal cells	Disorganised

🏆 Part 2: End-of-Chapter Master Lists (The Grand Finale) 🌻🐝

📚 1. Combined Terms List (Glossary) 🧠

Biological Term 🔑	Exam-Ready Definition 📝
Pollination	The process of transfer of pollen grains from the anther to the stigma of the same species.
Autogamy	Self-pollination where pollen falls on the stigma of the same flower.
Geitonogamy	Self-pollination where pollen falls on the stigma of another flower of the same plant.
Allogamy	Cross-pollination where pollen transfers to the stigma of another flower of a different plant of the same species.
Cleistogamous	Minute, bisexual flowers that do not open at all, ensuring self-pollination.
Chasmogamous	Large flowers with open petals and exposed reproductive structures that favor cross-pollination.
Homogamy	Condition where the anther and stigma of a flower mature at the same time.
Dichogamy	Condition where the anther and stigma of a flower mature at different times.
Protandry	A type of dichogamy where anthers mature earlier than the stigma.

Biological Term 🔑	Exam-Ready Definition 📝
Protogyny	A type of dichogamy where the stigma matures earlier than the anthers.
Herkogamy	Mechanical or structural barriers (like a hood) that prevent pollen from reaching the stigma of the same flower.
Heterostyly	Condition where the stigma and anthers grow at different heights, preventing self-pollination.
Entomophilous	Flowers that are pollinated by insects.
Anemophilous	Flowers that are pollinated by wind.
Hydrophilous	Flowers that are pollinated by water.
Ornithophily	Pollination facilitated by birds.
Chiropterophily	Pollination facilitated by bats.
Elephophily	Pollination facilitated by elephants.
Emasculation	The artificial removal of anthers in young flowers by plant breeders to prevent self-pollination.
Bagging	Covering emasculated flowers with plastic bags to prevent unwanted natural pollination.

Biological Term 🔑	Exam-Ready Definition 📝
Fertilisation	The union/fusion of the nuclei of male and female gametes.
Syngamy	The fusion of one sperm nucleus with the egg cell nucleus to form a zygote.
Triple Fusion	The fusion of the second sperm nucleus with the two polar nuclei to form the endosperm nucleus.
Double Fertilisation	The occurrence of two fertilizations (syngamy and triple fusion) simultaneously within the embryo sac.

🌼 2. Master Example List 🌿

- Flower Types by Pollination Mechanism:
  - *Cleistogamous (Closed/Self):* Pea, pansy.
  - *Chasmogamous (Open/Cross):* Oxalis, Hibiscus.
- Nature's Devices for Cross-Pollination:
  - *Unisexual (Separate trees):* Palms, papaya.
  - *Unisexual (Same plant):* Cucumber, gourd.
  - *Protandry (Anthers mature first):* Lady finger, sweet pea, Salvia, sunflower, maize.
  - *Protogyny (Stigma matures first):* Custard apple, peepal.
  - *Self-Sterility:* Ray florets of sunflower, orchids.
  - *Herkogamy (Mechanical barrier):* Pansy, Iris.
  - *Heterostyly (Different heights):* Primrose, Oxalis.
- Agents of Pollination:

- *Insect-Pollinated:* Dahlia, sweet pea, China rose, Salvia, sunflower.
- *Wind-Pollinated:* Maize, grass, rice, wheat.
- *Water-Pollinated:* Hydrilla, Vallisneria.
- *Bird-Pollinated:* Bignonia, Canna.
- *Elephant-Pollinated:* Rafflesia.

- Fate of Calyx (Sepals) Post-Fertilization:
  - *Dried and shrivelled form persists:* Apple, guava.
  - *Green calyx persists:* Brinjal.

⚙️ 3. Master Function & Location Matrix 📍🛠️

Biological Part 🧬	Exact Anatomical Location 📍	Primary Function / Post-Fertilization Fate ⚙️
Exine	Outer wall of the mature pollen grain.	Protective covering; has a point where the pollen tube grows out.
Intine	Inner wall of the mature pollen grain.	Protective inner covering.
Generative Nucleus	Inside the pollen grain/pollen tube.	Divides into two sperm nuclei (male gametes) for fertilization.
Tube Nucleus	At the tip of the growing pollen tube.	Directs the growth of the pollen tube all the way down, then disintegrates.

Biological Part 🧬	Exact Anatomical Location 📍	Primary Function / Post-Fertilization Fate ⚙️
Micropyle	A small opening at one end of the ovule.	Allows for the entry of the pollen tube.
Nucellus	Inside the ovule, enclosed by integuments.	Mass of food-laden cells.
Embryo Sac	Further inside the nucellus.	Contains 7 cells (including egg and polar nuclei) for fertilization.
Synergids	Micropylar end of the embryo sac (2 cells).	Help guide the pollen tube; disorganize after fertilization.
Antipodal Cells	Opposite end of the embryo sac (3 cells).	Disorganize after fertilization.
Ovary	Basal portion of the carpel.	Enlarges to form the Fruit.
Ovary Wall	Outer boundary of the ovary.	Becomes the Pericarp (Fruit wall).
Ovule	Inside the ovary cavity.	Becomes the Seed.

Biological Part 📌	Exact Anatomical Location 📍	Primary Function / Post-Fertilization Fate ⚙️
Placenta	Attaches ovule to ovary wall.	Becomes the Stalk of the seed.
Outer Integument	Outer protective covering of ovule.	Becomes the Testa (Outer seed coat).
Inner Integument	Inner protective covering of ovule.	Becomes the Tegmen (Inner seed coat).
Egg Cell	Inside the embryo sac.	Fuses with sperm nucleus to become the Zygote -> Embryo.
Polar Nuclei (Central Cell)	Center of the embryo sac.	Fuses with sperm nucleus to become the Endosperm.

C. SHORT ANSWER TYPE 📝

1. Explain briefly the following terms: 🧐

(a) Ornithophily: It is the pollination of flowers by birds (e.g., *Bignonia*, *Canna*). 🐓

(b) Elephophily: It is the pollination facilitated by elephants, as seen in *Rafflesia* where flowers are very large and pollen gets stuck to their feet. 🐘

(c) Artificial pollination: It is a standard practice where man himself transfers pollen to the stigma (also called artificial crossing) to evolve new plant varieties. 👩‍🌾

2. Define: 📖

(a) Pollination: The process of transfer of pollen grains from the anther to the stigma of a flower. 🌼

(b) Herkogamy: A condition in some flowers where mechanical or structural barriers prevent the pollen from reaching the stigma of the same flower (e.g., a hood covering the stigma in Pansy). 🛡️

(c) Cleistogamy: A condition where minute bisexual flowers do not open at all even at maturity, keeping reproductive structures hidden and forcing self-pollination (e.g., pea, pansy). 🔒

(d) Emasculation: The removal of anthers in young flowers during artificial pollination to prevent self-pollination. ✂️

3. What happens to the following after fertilization? ✨

(a) Ovules: Become the seeds. 🌱

(b) Calyx: May either fall off or remain intact in a dried and shrivelled form (and sometimes persists as green calyx, like in brinjal). 🍂

(c) Petals: Wither and generally fall off. 🥀

(d) Stamens: Wither and generally fall off. 💨

4. Mention any two contrivances in flowers which favour cross-pollination. 🌿

1. **Unisexuality:** Flowers are either male or female, making cross-pollination the only possibility. 🌺
2. **Dichogamy:** Different timings for maturation of androecium and gynoecium in the same flower. ⏱️

(Other acceptable answers include Self-sterility, Herkogamy, or Heterostyly).

D. LONG ANSWER TYPE ✍️

1. Distinguish between the following pairs: 🔍

(a) Autogamy and Geitonogamy:

- **Autogamy:** Pollen of the *same flower* falls on its own stigma. 🔄
- **Geitonogamy:** Pollen of *another flower* of the *same plant* falls on the stigma. 🌿

(b) Homogamy and Dichogamy:

- **Homogamy:** The anther and stigma of a flower mature at the *same time* to ensure self-pollination. ⏰
- **Dichogamy:** The anther and stigma mature at *different times*, favouring cross-pollination. ⏳

(c) Protandry and Protogyny:

- **Protandry:** Anthers of the flower mature *earlier* than the stigma (e.g., sweet pea). 🟡
- **Protogyny:** Stigma of the flower matures *earlier* than the anthers (e.g., custard apple). 🎯

(d) Entomophilous and Anemophilous flowers:

- **Entomophilous (Insect-pollinated):** Large, brightly coloured, emit scent, produce nectar, with sticky or spiny pollen. 🦋
- **Anemophilous (Wind-pollinated):** Small, not brightly coloured, no scent or nectar, with large quantities of light, dry, smooth pollen. 🌬️

(e) Advantages of self and cross-pollination:

- **Self-pollination:** Surer process, preserves parental characters indefinitely, highly economical (no need for large petals, scent, or excess pollen). 💰
- **Cross-pollination:** Offspring are healthier, produces abundant and viable seeds, and yields new varieties. 🌟

2. What are the advantages of the following in the flower to the plant concerned? 💡

(a) Long and feathery stigma: Helps to easily trap wind-blown pollen grains in wind-pollinated flowers. 🪶

(b) Brightly coloured petals: Attracts insects for cross-pollination. 🦋

(c) Smooth and light pollen: Allows the pollen to be easily carried away by wind over long distances. 🌬️

(d) Protruding and easily movable anthers: So that the slightest wind can move them (versatile anthers) to easily blow the pollen away. 🌾

(e) Fragrant nectar: Acts as a food source to attract insects, ensuring they visit and pollinate the flower. 🍯

3. Describe the advantages and disadvantages of cross-pollination to the plant. ⚖️

- **Advantages:** The offspring produced are healthier and can better adapt to environmental changes. The seeds produced are abundant and highly viable. It also allows for the creation of new varieties. 🧬
- **Disadvantages:** It is not always certain because it relies on external pollinating agents. Large quantities of pollen must be produced, leading to wastage. It is also uneconomical for the plant as it must expend energy to produce large, scented, nectar-filled flowers. 📉

4. What is the function of the pollen tube? Explain it with the help of a diagram. 🔬

(Diagram omitted as per your specific instructions!) 🚫🖍️

- **Function:** The pollen tube grows through the stigma and style by dissolving tissues to reach the ovary. It acts as a carrier, pushing through the micropyle to deliver the two sperm nuclei (male gametes) directly into the embryo sac to complete double fertilization. 🚀

E. STRUCTURED / APPLICATION / SKILL TYPE 🧩

1. The figure shown below represents a process in flowers. Study the same and answer the following questions.

(a) Name and define the process.

- **Process:** Cross-pollination. 🌼
- **Definition:** It is the transfer of pollen grains from the anthers of a flower of one plant to the stigma of a flower of another plant of the same species. 🐝

(b) Write the technical term of the above mentioned process.

- **Allogamy.** 🌿

(c) Give two examples of plants in which this process occurs.

- *Oxalis* and *Hibiscus*. 🌺

(d) Write one advantage and one disadvantage of the process shown.

- **Advantage:** Offspring are healthier. 🌟
- **Disadvantage:** Pollination is not always certain as it depends on external agents. 🌧️

(e) Name two agents that participate in this process.

- Insects and Wind. 🦋 🌬️

2. Given alongside is a diagrammatic sketch of the sectional view of a germinating pollen grain. Study the same and then answer the questions that follow:

(a) Name the parts labelled 1, 2, 3, 4 and 5.

1. Exine (outer wall) 🛡️
2. Intine (inner wall) 🟢
3. Tube nucleus 🍼
4. Pollen tube 🏞️
5. Generative nucleus (or Sperm nuclei) 🧬

(b) Where does the germination of the pollen grain takes place and how?

- It takes place on the **stigma** of a flower of the same species. It is stimulated to germinate by the secretion of sugars by the stigma, causing the pollen tube to grow out through a germ pore in the exine. 💧

(c) What is the function of the part labelled '4'?

- The pollen tube (4) carries the male gametes (sperm nuclei) down through the style and into the ovule for fertilization. 🚀

(d) What happens to the part labelled '5' during the process?

- The generative nucleus (5) divides into **two sperm nuclei** (male gametes) as it travels down the pollen tube. ➗

3. Given below is a diagrammatic representation of the process of fertilization. Study the same and then answer the questions that follow:

(a) Name the parts labelled 1, 2, 3, 4, 5 and 6.

1. Pollen grain 🟡
2. Pollen tube 🏞️
3. Polar nuclei (in the central cell) 🔮
4. Embryo sac 🛸
5. Antipodal cells 🔬
6. Micropyle 🕳️

(b) What happens to (i) Ovary (ii) Ovule after fertilization?

- (i) The **Ovary** enlarges to form the **fruit**. 🍎
- (ii) The **Ovules** become the **seeds**. 🌱

(c) What is the function of the synergids?

- They help to assist and guide the sperm nucleus into the egg cell during fertilization. 🧭

(d) What part does the stigma play in the process of fertilization?

- It acts as the landing place for pollen grains and secretes sugars that stimulate the pollen to germinate and grow a pollen tube, initiating the path to fertilization. 🎯

4. Shivam dissected a flower of sweet pea and removed all the petals. He observed the reproductive parts of the flower (both male and female), arranged in a specific manner as shown in the diagram given below.

(a) Mention the kind of arrangement of the stamens as shown in the figure.

- **Diadelphous** (stamens are united in two bundles; nine form a staminal tube while one is free). 🫛

(b) What are the three kinds of petals that Shivam removed from the flower? Name them.

- Standard petal, Wing petals, and Keel petals. 🌸

(c) Name the kind of pollination (on the basis of pollinating agent) with one specific feature of the flower which favours it.

- **Kind of pollination:** Insect-pollinated (Entomophily). 🦋
- **Specific feature:** The flower has brightly coloured, conspicuous corolla (petals) to attract insects like bees, and a mechanism where the wing and keel petals are forced down by the insect's weight to expose the stamens and stigma. 🌺

F. ASSERTION - REASON TYPE QUESTIONS ⚖️

(Choose the answer based on the codes given below:)

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

(i) Assertion (A): Autogamy and allogamy are the two kinds of self-pollination. Reason (R): Self-pollination occurs within the same flower or two flowers of the same plant.

- **Answer: (d)** A is false but R is true. (Assertion is false because *allogamy* is cross-pollination, not self-pollination. Geitonogamy is the other type of self-pollination). ❌

(ii) Assertion (A): Herkogamous flowers favour cross-pollination as these flowers have a structural barrier between the male and female reproductive parts. Reason (R): Pansy is an example of a herkogamous flower which has a hood covering the stigma.

- **Answer: (b)** Both A and R are true but R is not the correct explanation of A. (The reason is just an *example* of the assertion, not the explanation of the fundamental concept). 🛡️

(iii) Assertion (A): Entomophilous flowers are brightly coloured and often scented. Reason (R): The pollinating agents of entomophilous flowers are insects which usually get attracted towards the bright colours or scents of the flowers.

- **Answer: (a)** Both A and R are true and R is the correct explanation of A. 🦋

(iv) Assertion (A): One polar nucleus fuses with two sperm nuclei of a flower and develop into an endosperm during the process of fertilization. Reason (R): The large central cell of the embryo sac contains two polar nuclei, which fuse with one male nucleus. This process is termed as 'triple fusion'.

- **Answer: (d)** A is false but R is true. (Assertion is false because it is *two* polar nuclei fusing with *one* sperm nucleus, not the other way around). 🧬

(v) Assertion (A): The process of removal of anthers in young flowers is termed as 'emasculation'. Reason (R): Emasculation is mainly performed during artificial pollination to produce the desired variety of the plants.

- **Answer: (b)** Both A and R are true but R is not the correct explanation of A. (The reason explains *why* we do it, but it does not technically explain the definition/mechanism stated in the assertion). ✂️

G. CASE STUDY BASED QUESTION 📖

1. Shaista is working on her school project about plant reproduction. Her teacher Mrs. Aggrawal gave her the task of investigating pollination in pea flower, a common garden plant. Shaista is curious to know how pollination in pea flower and maize flower differs from each other.

(a) Is pollination a necessary event before fertilization for the reproduction in plants? Explain.

- **Yes.** Pollination is absolutely necessary because it is the process that physically transports the male gametes (pollen grains) from the anther to the female reproductive part (stigma). Without this transfer, the male gametes would never reach the ovule, making fertilization impossible. 🚀

(b) What are the pollinating agents for pea and maize flowers respectively?

- **Pea:** Insects. 🦋
- **Maize:** Wind. 🌬️

(c) Write scientific terms for the kind of pollination found in pea flower and maize flower on the basis of pollinating agents.

- **Pea flower:** Entomophily (Insect-pollinated). 🐝

- **Maize flower:** Anemophily (Wind-pollinated). 🌾

(d) Mention the technical term for the specific condition of stamens in pea flower.

- **Diadelphous** (Stamens are united in two bundles). 🫛

(e) What are the two advantages of cross-pollination?

- 1. The offspring produced are healthier and better adapted. 💪
- 2. It introduces genetic variation, allowing for new varieties to be produced. 🌟